12TH CHEMISTRY · CHAPTER 25 · FULL CHAPTER NOTES

# Organic Synthesis

Topic-wise short and long questions with answers from the complete chapter

## INTRODUCTION

### Q.1 [Long] What is the role of functional groups in organic synthesis?

The study of organic chemistry is organised around functional groups. Although millions of organic compounds are known, there are only a handful of functional groups, and each one defines a family of organic compounds. A functional group is the part of an organic compound where most of the reactions occur.

In organic synthesis we design different possible ways to build a target molecule from simpler, easily available components. Functional groups play a central role in designing a synthesis: they enable chemists to predict and control the reactions which ultimately lead to the desired product. Identifying the functional group present helps to predict the properties of a compound. This information is crucial for the development of new organic products, and to ensure the purity and composition of products such as medicines and food additives.

### Q.2 [Short] What is a functional group? Why is it important to identify it?

A functional group is the part of an organic compound where most of the reactions occur; each one defines a family of organic compounds. Identifying it helps to predict the properties and reactions of the compound, which is crucial for developing new organic products and for ensuring the purity and composition of products such as medicines and food additives.

## 25.1 SYNTHESIS AND RETROSYNTHESIS

### Q.3 [Short] What are the features of a good synthesis?

Before preparing an organic compound, it is important to understand the chemical reactions, the reagents and the conditions required to complete each step. Most organic compounds can be prepared by a number of different routes, so criteria are required to select the best route. A good synthesis converts cheap and easily available starting materials into the desired product by a route which involves the least number of steps and gives a high overall yield.

### Q.4 [Short] Into which two categories are the reactions used in organic synthesis divided? Why is carbon–carbon bond formation important?

The reactions used in organic synthesis are categorised by whether they bring about **functional group interconversions (FGI)** or **carbon–carbon bond formation**. Carbon–carbon bond formation reactions are particularly important, because the basic strategy of synthesis is to assemble the target molecule from simple and smaller starting materials.

### Q.5 [Long] What is synthesis? Explain with the example of a primary aliphatic amine.

In synthesis, the target molecule is prepared starting from simple, commercially available compounds. It may involve many steps or transformations to reach the final target molecule. For example, a primary aliphatic amine can be prepared in a single step by treating a haloalkane with ammonia:

$$R\text{–}CH_2\text{–}Cl + NH_3 \rightarrow R\text{–}CH_2\text{–}NH_2 + HCl$$

Since the haloalkane is not easily available, it is prepared from an alcohol, which is a more easily available starting material. The synthesis of the alkyl amine is then completed in two steps:

$$R\text{–}CH_2\text{–}OH \rightarrow R\text{–}CH_2\text{–}Cl\ (SOCl_2)$$
$$R\text{–}CH_2\text{–}Cl + NH_3 \rightarrow R\text{–}CH_2\text{–}NH_2 + HCl$$

### Q.6 [Short] Why is the synthesis of an alkyl amine carried out in two steps starting from an alcohol?

The amine is made from a haloalkane and ammonia, but the haloalkane is not easily available. An alcohol is a more easily available starting material, so it is first converted into the haloalkane with $SOCl_2$, and this is then treated with $NH_3$.

### Q.7 [Long] What is retrosynthesis? What is the disconnection approach?

Retrosynthesis is exactly the opposite of synthesis. It involves the breaking down of the target molecule into less complicated, simpler and more readily available compounds which can serve as starting materials. It uses the **disconnection approach**: those bonds of the target molecule are disconnected which result in a considerable structural simplification. Following a series of disconnections, we eventually arrive at simple, easily available starting materials.

Retrosynthesis: Target molecule $\Longrightarrow$ Starting material

Synthesis: Starting material $\rightarrow$ Target molecule

### Q.8 [Long] Differentiate between synthesis and retrosynthesis.

| Synthesis | Retrosynthesis |
|---|---|
| The target molecule is prepared starting from simple, commercially available compounds. | The target molecule is broken down (on paper) into simpler, readily available compounds. |
| It moves forward: starting material → target molecule. | It moves backward: target molecule ⟹ starting material. |
| It consists of actual reactions with reagents and conditions. | It consists of imaginary disconnections and functional group interconversions. |
| It is shown by the ordinary reaction arrow (→). | It is shown by the open-ended retrosynthetic arrow (⟹). |

### Q.9 [Long] Explain the terms used in retrosynthesis.

| Term | Meaning |
|---|---|
| Retrosynthetic arrow | An open-ended arrow, denoted by ⟹. |
| Target molecule (TM) | The molecule which is to be prepared by disconnection. |
| Disconnection | A process which involves the imaginary breaking down of the TM progressively into simpler starting materials. |
| Synthons | Fragments of the target molecule which are generated by a disconnection. They are often ionic species and represent the building blocks for the synthesis. |
| Synthetic equivalent (SE) | A reagent used to carry out the function of a synthon, which cannot be used itself because it is usually unstable. |
| Functional group interconversion (FGI) | The conversion of one functional group into another to reach the target molecule. |
| Retrosynthetic tree | A diagram showing all possible disconnections and synthons from the target molecule to the starting materials. |

### Q.10 [Short] What are synthons? Why are synthetic equivalents needed?

Synthons are the fragments of the target molecule generated by a disconnection; they are often ionic species and represent the building blocks of the synthesis. A synthon is usually unstable and cannot be used itself, so a real reagent, the synthetic equivalent, is used to carry out its function.

### Q.11 [Short] Explain the terms of retrosynthesis using aniline as the target molecule.

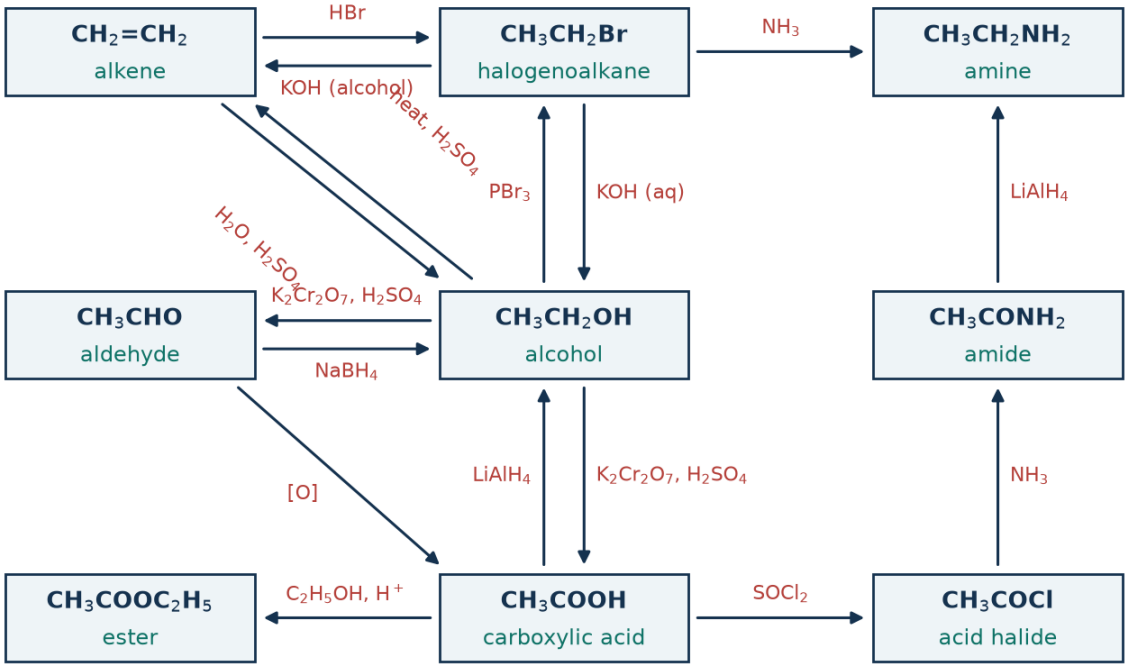
The target molecule (TM) is aniline. By FGI the $-NH_2$ group is changed to $-NO_2$, giving nitrobenzene. Disconnection of its C–N bond gives synthon A, the aryl anion ($C_6H_5^-$), whose synthetic equivalent is benzene, and synthon B, $NO_2^+$, whose synthetic equivalent is nitric acid.

## Q.12 [Long] What is functional group interconversion (FGI)? What is its role in designing a synthesis?

FGI is the change of one functional group into another functional group, required for the synthesis of a specific compound. It has a crucial role in designing routes which lead to target molecules. Simple starting materials can be converted into more complex target molecules using FGI as building blocks. FGI not only allows the carbon skeleton of the target molecule to be created, but also selects the route to be adopted for this purpose.

## Q.13 [Long] Summarise the functional group interconversions of aliphatic compounds.

The interconversions studied in earlier chapters are summarised below.




*Functional group interconversion in aliphatic organic compounds*

| Conversion | Reagent(s) |
|---|---|
| Alkene → halogenoalkane | HBr |
| Halogenoalkane → alkene | KOH (alcoholic) |
| Alkene → alcohol | $H_2O$, $H_2SO_4$ |
| Alcohol → alkene | heat, $H_2SO_4$ |
| Halogenoalkane → alcohol | KOH (aq) |
| Alcohol → halogenoalkane | $PBr_3$ |
| Halogenoalkane → amine | $NH_3$ |
| Alcohol → aldehyde → carboxylic acid | $K_2Cr_2O_7$, $H_2SO_4$ |
| Aldehyde → alcohol | $NaBH_4$ |

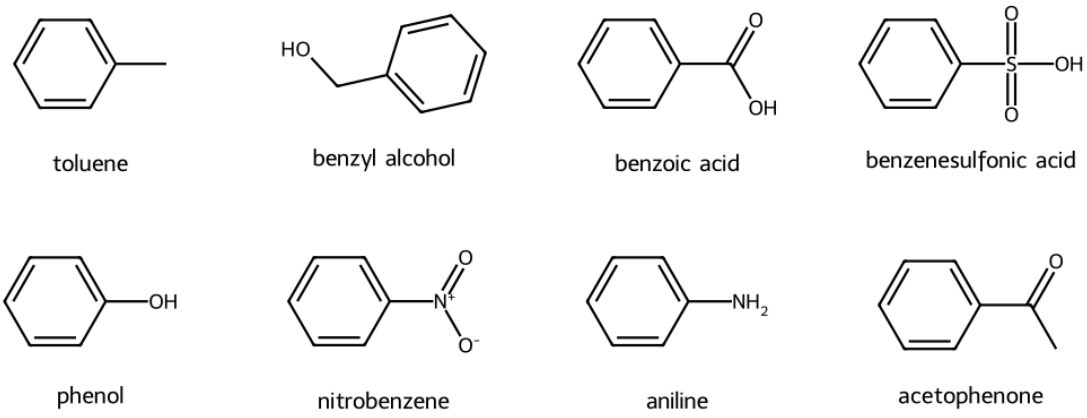
| Conversion | Reagent(s) |
|---|---|
| Carboxylic acid → alcohol | $LiAlH_4$ |
| Carboxylic acid → ester | $C_2H_5OH$, $H^+$ |
| Carboxylic acid → acid halide | $SOCl_2$ |
| Acid halide → amide | $NH_3$ |
| Amide → amine | $LiAlH_4$ |

*In Figure 25.3 the textbook labels the aldehyde → carboxylic acid arrow "$V_2O_5$" and also draws $NH_3$ arrows from the alcohol to the amine and amide; these are not reactions described in the earlier chapters and are not included here.*

## Q.14 [Long] Summarise the functional group interconversions of aromatic compounds starting from benzene.

Benzene is converted into the following compounds:

| Conversion | Reagent(s) |
|---|---|
| Benzene → toluene | $CH_3Cl$, $AlCl_3$ |
| Toluene → benzyl alcohol | (i) $Cl_2$, light (ii) $KOH_{(aq)}$ |
| Toluene → benzoic acid | $KMnO_4$, $H_2SO_4$ |
| Benzene → benzenesulfonic acid | $SO_3$, $H_2SO_4$ |
| Benzenesulfonic acid → phenol | (i) NaOH (ii) HCl |
| Benzene → nitrobenzene | $HNO_3$, $H_2SO_4$ |
| Nitrobenzene → aniline | Sn, HCl |
| Benzene → acetophenone | $CH_3COCl$, $AlCl_3$ |




*Aromatic compounds obtained from benzene by functional group interconversion*

## Quick Check 25.1

### Q.15 [Short] i. Why aldehydes and ketones undergo nucleophilic addition reaction readily?

The carbonyl group is polar: the carbon atom carries a partial positive charge (δ+) and is electrophilic, so it is easily attacked by nucleophiles. The π bond of the C=O group is easily broken, and the planar shape of the group leaves the carbonyl carbon uncrowded. Hence addition of nucleophiles takes place readily.

### Q.16 [Short] ii. Convert a carboxylic acid to its anhydride.

The acid is first converted into its acid chloride (an FGI), which is then treated with the sodium salt of the same acid:

$$CH_3\text{–}COOH + SOCl_2 \rightarrow CH_3\text{–}COCl + SO_2 + HCl$$
$$CH_3\text{–}COCl + CH_3\text{–}COONa \rightarrow (CH_3CO)_2O + NaCl$$

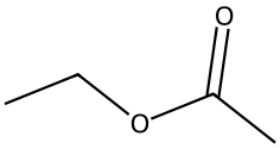
*The preparation of acid anhydrides is not described in the textbook; this is the standard method. (Acetic anhydride appears in the book only as a reagent for aspirin.)*

## 25.2 RETROSYNTHESIS AND SYNTHESIS OF COMPOUNDS

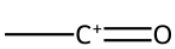
### Q.17 [Long] Construct the retrosynthetic tree of ethyl acetate and design its synthesis.

**Step I – Disconnection:** the C–O bond of the ester group is disconnected. This gives synthon A (an acyl cation) and synthon B (the ethoxide ion). While selecting the synthetic equivalents, they should undergo a reaction to give the target molecule.

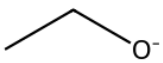
**Step II – Synthetic equivalents:** the synthons are neutralised to get the synthetic equivalents, acetic acid and ethanol.



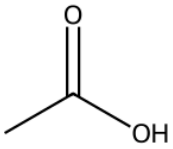
TM: ethyl acetate



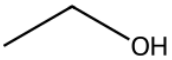
synthon A



synthon B



SE of A: acetic acid



SE of B: ethanol

**Synthesis:** the familiar esterification reaction between acetic acid and ethanol.

$$CH_3\text{–}COOH + C_2H_5\text{–}OH \rightarrow CH_3\text{–}COO\text{–}C_2H_5 + H_2O\ (H_2SO_4)$$

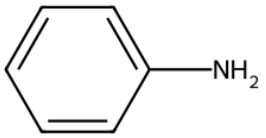
*In Step II the textbook sentence says the synthetic equivalents are "benzene and nitric acid"; its own figure and the next sentence show that they are acetic acid and ethanol.*

### Q.18 [Long] Construct the retrosynthetic tree of aniline and design its synthesis.

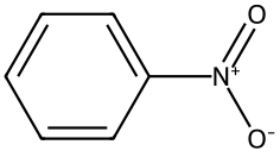
**Step I – Functional group interconversion:** the amino group (–$NH_2$) is converted into the nitro group (–$NO_2$), because it is easy to attach a nitro group to benzene by electrophilic substitution, rather than an amino group by nucleophilic substitution.

**Step II – Disconnection:** nitrobenzene is disconnected at the C–N bond, which creates the synthons aryl anion ($C_6H_5^-$) and nitronium cation ($NO_2^+$).

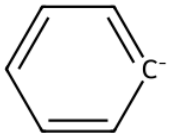
**Step III – Synthetic equivalents:** the synthons are neutralised to get benzene and nitric acid.




TM: aniline




after FGI: nitrobenzene




synthon A: aryl anion




synthon B: nitronium ion



SE of A: benzene (B: nitric acid)

**Synthesis:**

$$C_6H_6 + HNO_3 \rightarrow C_6H_5\text{–}NO_2 + H_2O\ (H_2SO_4)$$
$$C_6H_5\text{–}NO_2 + 6[H] \rightarrow C_6H_5\text{–}NH_2 + 2H_2O\ (Sn + HCl)$$

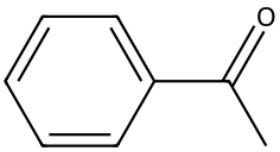
### Q.19 [Short] Why is an FGI the first step in the retrosynthesis of aniline?

Because it is easy to attach a nitro group to benzene by electrophilic substitution, whereas an amino group cannot be attached directly by nucleophilic substitution. So the amino group is first changed (on paper) into a nitro group.

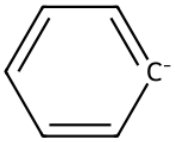
### Q.20 [Long] Construct the retrosynthetic tree of acetophenone and design its synthesis.

**Step I – Disconnection:** acetophenone is disconnected at the C–C bond between the ring and the carbonyl group, which creates the synthons aryl anion ($C_6H_5^-$) and acylium cation ($CH_3CO^+$).

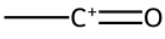
**Step II – Synthetic equivalents:** the synthons are neutralised to get benzene and acyl chloride.




TM: acetophenone



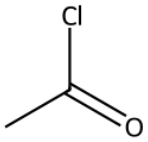

synthon A: aryl anion




synthon B: acylium ion



SE of A: benzene




SE of B: acyl chloride

**Synthesis:** benzene reacts with the acyl chloride in the presence of $AlCl_3$ as catalyst (Friedel–Crafts acylation).

$$C_6H_6 + CH_3\text{–}COCl \rightarrow C_6H_5\text{–}CO\text{–}CH_3 + HCl\ (AlCl_3)$$

### Q.21 [Short] Who first synthesised cholesterol?

The landmark synthesis of cholesterol was first achieved in the early 1950s by two competing groups, headed by Robert Robinson at Oxford and R.B. Woodward at Harvard. These synthetic schemes of cholesterol had previously been predicted through biological studies.

## Quick Check 25.2

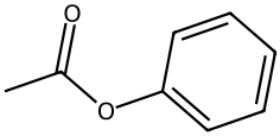
### Q.22 [Short] a) Why nitration of benzene is done during synthesis of aniline?

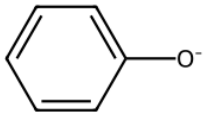
It is easy to attach a nitro group to the benzene ring by electrophilic substitution, whereas an amino group cannot be attached directly by nucleophilic substitution. So benzene is first nitrated, and the nitro group is then reduced (Sn + HCl) to the amino group.

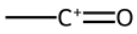
### Q.23 [Short] b) Phenyl ethanoate is an important molecule used in synthesis of medicines for hepatitis and various infection. i) Draw its retrosynthetic tree. ii) Write structures of its synthetic equivalents. iii) Design its linear synthesis.

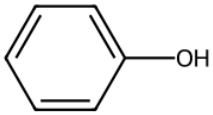
**i) Retrosynthetic tree:** the C–O bond of the ester linkage is disconnected, giving synthon A (phenoxide ion) and synthon B (acylium ion).

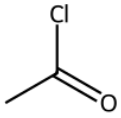
**ii) Synthetic equivalents:** phenol, $C_6H_5$–OH, for synthon A and ethanoyl (acyl) chloride, $CH_3$–COCl, for synthon B.


TM: phenyl ethanoate


synthon A: phenoxide ion


synthon B: acylium ion


SE of A: phenol


SE of B: ethanoyl chloride

**iii) Linear synthesis** (from benzene, using the conversions of Figure 25.4):

$$C_6H_6 \rightarrow C_6H_5\text{–}SO_3H\ (SO_3,\ H_2SO_4)$$
$$C_6H_5\text{–}SO_3H \rightarrow C_6H_5\text{–}OH\ ((i)\ NaOH,\ (ii)\ HCl)$$
$$C_6H_5\text{–}OH + CH_3\text{–}COCl \rightarrow CH_3\text{–}COO\text{–}C_6H_5 + HCl$$

## 25.3 SYNTHESIS OF PHARMACEUTICAL PRODUCTS

### Q.24 [Short] Why is retrosynthesis important in the synthesis of pharmaceutical products?

One of the most important applications of retrosynthesis is in the synthesis of pharmaceutical products. There are hundreds of valuable medicinal compounds which look very complex in structure, but if they are disconnected properly and efficiently, we can approach very simple and commonly available starting materials. This is the beauty of organic synthesis: it makes a highly complex and terrifying structure look simple and achievable after a series of efficient retrosynthetic steps.

### Q.25 [Long] Construct the retrosynthetic tree of paracetamol and design its synthesis from phenol.

The direct synthesis of paracetamol from phenol requires a series of steps, which can be decoded by retrosynthesis.

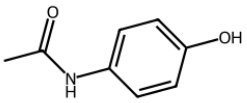
**Step I – Disconnection:** paracetamol is disconnected at the C–N bond of the amide group to get the synthons.

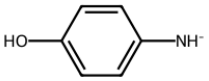
**Step II – Synthetic equivalents:** 4-aminophenol and acyl chloride. The bond is broken at this position because the synthetic equivalents are well known to react through a nucleophilic addition–elimination reaction.

**Step III – Functional group interconversion:** 4-aminophenol $\Longrightarrow$ 4-nitrophenol, because it is easy to prepare p-nitrophenol, which can then be reduced to p-aminophenol.

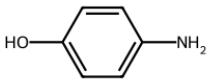
**Step IV – Disconnection:** 4-nitrophenol is disconnected at the C–N bond, creating the 4-hydroxy aryl anion and the nitronium cation ($NO_2^+$).

**Step V – Synthetic equivalents:** phenol and nitric acid.

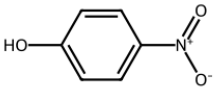

TM: paracetamol

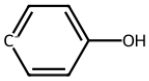

synthon A


synthon B

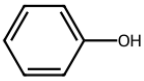

SE of A: 4-aminophenol


SE of B: acyl chloride


after FGI: 4-nitrophenol


synthon A: 4-hydroxy aryl anion


synthon B: nitronium ion


SE of A: phenol (B: nitric acid)

**Synthesis of paracetamol** from phenol (starting material):

$$HO–C_6H_5 + HNO_3 \rightarrow HO–C_6H_4–NO_2 + H_2O\ (H_2SO_4)$$

$$HO–C_6H_4–NO_2 + 6[H] \rightarrow HO–C_6H_4–NH_2 + 2H_2O\ (Sn + HCl)$$

$$HO–C_6H_4–NH_2 + CH_3–COCl \rightarrow HO–C_6H_4–NH–CO–CH_3 + HCl$$

*In Step V the textbook sentence names the synthetic equivalents as “benzene and nitric acid”; its figure shows phenol and nitric acid. “4-hyrodxy aryl anion” in Step IV is a misprint of 4-hydroxy aryl anion.*

### Q.26 [Short] Why is paracetamol disconnected at the C–N bond of the amide group?

Because the synthetic equivalents of the resulting synthons, 4-aminophenol and acyl chloride, are well known to react through a nucleophilic addition–elimination reaction to give the amide.

### Q.27 [Long] Construct the retrosynthetic tree of aspirin and design its synthesis from phenol.

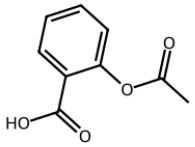
Like paracetamol, the direct synthesis of aspirin (acetylsalicylic acid) requires a series of steps. Retrosynthesis shows how many and which steps are needed to prepare aspirin from phenol (a simple starting material).

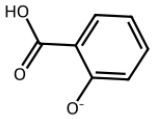
**Step I – Disconnection-I:** aspirin is disconnected at the C–O bond of the ester linkage to get the synthons.

**Step II – Synthetic equivalents:** salicylic acid and acetic anhydride.

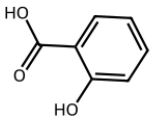
**Step III – Disconnection-II:** the C–C bond of the carboxylic group of salicylic acid is disconnected.

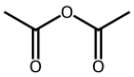
**Step IV – Synthetic equivalents:** phenol and carbon dioxide.

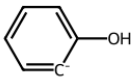

TM: aspirin


synthon A

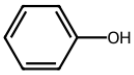

synthon B


SE of A: salicylic acid


SE of B: acetic anhydride


synthon A (from salicylic acid)


synthon B


SE of A: phenol (B: carbon dioxide)

**Synthesis of aspirin:**

$$C_6H_5\text{–}OH + CO_2 \rightarrow HO\text{–}C_6H_4\text{–}COOH \text{ ((i) high T and P; (ii) } H_2SO_4)$$
$$HO\text{–}C_6H_4\text{–}COOH + (CH_3CO)_2O \rightarrow CH_3COO\text{–}C_6H_4\text{–}COOH + CH_3COOH \text{ } (H_3PO_4)$$

Phenol is first converted into salicylic acid, which is then heated with acetic anhydride and $H_3PO_4$ to give aspirin (acetic acid is the by-product).

**Q.28** [Short] **Name the synthetic equivalents obtained in the two disconnections of aspirin.**

Disconnection-I (C–O bond of the ester linkage): salicylic acid and acetic anhydride. Disconnection-II (C–C bond of the carboxylic group): phenol and carbon dioxide.

## Quick Check 25.3

**Q.29** [Short] **Aspirin and Paracetamol, both are synthesized starting from phenol. I) What is the overall functional group difference in them? ii) In synthesis of which FGI (Functional Group Interconversion) is involved? iii) Which synthon is common in preparation of them? Write down its structure and also mention its most common synthetic equivalent.**

**I)** Aspirin contains an **ester** group and a **carboxylic acid** group on the benzene ring. Paracetamol contains an **amide** group and a phenolic **hydroxyl** group.

**ii)** In the synthesis of **paracetamol**: the amino group of 4-aminophenol is obtained from the nitro group of 4-nitrophenol (reduction with Sn + HCl).

**iii)** The acylium (acetyl) cation, $CH_3\text{–}C^+{=}O$, is the common synthon. Its most common synthetic equivalent is acyl chloride (ethanoyl chloride, $CH_3COCl$); in the aspirin synthesis acetic anhydride is used.

## 25.4 USE OF AI IN DRUG DESIGNING

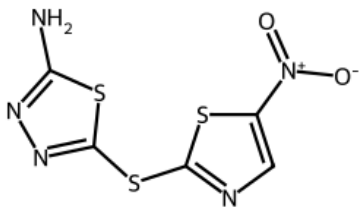
### Q.30 [Short] Why are new methods needed for developing antibiotics?

The past few decades have seen the development of very few new antibiotics, and those developed are only slightly different from the existing ones. The methods currently used to develop new drugs are very costly, time consuming and usually limited to a narrow spectrum of chemical activity. We are also facing a growing number of cases in which the prevalent antibiotics become resistant to existing pathogens.

### Q.31 [Long] How was Halicin discovered with the help of artificial intelligence?

To solve this problem, researchers at MIT (Massachusetts Institute of Technology) designed a machine-learning algorithm to look for chemical features that make molecules effective at killing *E. coli* bacteria. They applied the model to 2500 molecules and a set of 800 natural products with different structures and a wide range of bioactivities. The model picked out one molecule that was predicted to have antibacterial properties and a molecular structure different from the existing antibiotics. Using a different model, they also showed that this molecule would most likely have low toxicity to human cells. It was named **Halicin**.

Halicin was tested against a number of bacterial strains isolated from patients. It killed many of the world's most problematic disease-causing bacteria which were previously resistant to treatment.



Halicin

### Q.32 [Short] What did the researchers do after identifying Halicin? What is the importance of this work?

After identifying Halicin, the researchers used their computer model to screen more than 100 million molecules selected from a database. This screening was completed in only three days, and it selected 23 more molecules with structures different from existing antibiotics. Eight of these showed antibacterial activity and two were particularly powerful. This groundbreaking work points out an altogether different route to discover new organic molecules which may be used as drugs.

### Q.33 [Short] What is Halicin? Against which organisms is it effective?

Halicin is a molecule picked out by a machine-learning model at MIT as a new antibiotic with a structure different from existing antibiotics and with low predicted toxicity to human cells. It killed many of the world's most problematic disease-causing bacteria which were previously resistant to treatment.

### Q.34 [Short] Write a note on the total synthesis of vitamin $B_{12}$.

The historic total synthesis of vitamin $B_{12}$ was achieved by R.B. Woodward and A. Eschenmoser in 1972, after over 10 years of collaboration. The total synthesis involved over 100 steps and required 91 post-doctoral researchers and 12 PhD students.

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